

Assembly and Concentration at a Fixed-Shift Type-II Gate: Cumulative Divisor Prefixes, Bilinear Distribution Frontiers, and TT^* Dispersion

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Abstract

Fix $h \neq 0$, let $W \in C_c^\infty((0, \infty); \mathbb{R})$, and write

$$\Lambda_R(n) = \sum_{q \leq R} \frac{\mu(q)}{\varphi(q)} c_q(n), \quad r_R = \Lambda - \Lambda_R, \quad R = X^{1/2-\delta}.$$

TPC-16 isolated a three-scale fixed-shift Type-II packet with source scale L , quotient scale K , and divisor label D , and proved cancellation for one D -resolved packet near the $11/21$ factorable boundary. We replace that isolated slice by an assembled divisor prefix and then sharpen the spectral description of what remains.

The published core uses Maynard's 2025 fixed-residue distribution theorem. For a dyadic source block, all divisor labels $d \leq D_0$ are summed at once through

$$\beta_{\leq D_0}(k) = \sum_{\substack{d|k \\ d \leq D_0}} \mu(d).$$

Under the three Maynard monomial conditions, the corresponding residual packet has arbitrary logarithmic saving. In particular, the whole prefix $d \leq X^{1/21-o(1)}$ is packaged directly near $L = X^{10/21-o(1)}$, yielding an exact tail normal form.

An explicitly segregated extension imports Theorem 1.1 of Li's currently unrefereed arXiv:2602.20917v6, revised 27 May 2026. Its signed bilinear estimate deletes the third Maynard monomial condition. The resulting exponent polytope has vertex

$$(\log_X L, \log_X D_0) = \left(\frac{8}{17}, \frac{1}{17} \right), \quad \log_X(LD_0) = \frac{9}{17}.$$

Thus, within the TPC-16 interface and using that version-locked imported theorem, an entire prefix $d \leq X^{1/17+o(1)}$ is removed near $L = X^{8/17-o(1)}$. The proof records why the target weight depends on the source factor but not on d ; arbitrary pair-dependent masks are not covered. The published Maynard layer remains intact under any later revision of the preprint.

For the surviving dyadic block

$$\mathcal{H}_{L,K}(\alpha, \beta) = \sum_{\ell, k} c_{\ell, k} e(\alpha\ell + \beta k), \quad LK \asymp X,$$

we prove the explicit energy bound

$$\sum_{\ell, k} |c_{\ell, k}|^2 \ll X(\log X)^3.$$

Exact Nyquist Parseval makes every natural-size exceptional phase a finite translation-spectral atom. A flat-top reproducing formula shows that failure at the distinguished zero phase forces quantitative energy inside a polylogarithmically enlarged uncertainty cell. Finally, TT^* reduces such failure to a positive off-diagonal correlation of the residuals $r_R(\ell_1 k + h)$ and $r_R(\ell_2 k + h)$ of size $\gg XL/(\log X)^2$ at a dyadic witness scale. A sharp abstract countermodel shows that band limitation, total residual energy, and phase mean squares alone cannot exclude the zero mode.

No fixed-shift prime-pair asymptotic, twin-prime lower bound, new distribution theorem, or breach of sieve parity is claimed.

Keywords: fixed-shift prime correlations; factorable moduli; cumulative divisor prefixes; Ramanujan residuals; Type-II sums; phase concentration; TT^* dispersion; sieve parity.

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1 Introduction and theorem ledger

For fixed $h \neq 0$, consider the smoothed correlation

$$\mathcal{P}_h(X; W) = \sum_{n \geq 1} \Lambda(n) \Lambda(n+h) W(n/X). \quad (1)$$

Its Hardy–Littlewood main term is

$$\mathfrak{S}(h) X I_0(W), \quad I_0(W) = \int_0^\infty W(t) dt, \quad (2)$$

where

$$\mathfrak{S}(h) = \prod_p \frac{1 - \nu_p(h)/p}{(1 - 1/p)^2}, \quad \nu_p(h) = \#\{a \pmod p : a(a+h) \equiv 0 \pmod p\}. \quad (3)$$

For an odd h the factor at 2 vanishes. For a nonzero even h , (2) with a nonnegative nonzero W would imply infinitely many prime pairs at distance h . Nothing in this paper proves that asymptotic; the case $h = 2$ remains open [1, 2, 7].

TPC-15 introduced the finite Ramanujan model and reduced (1) to a signed Vaughan Type-II packet [8]. TPC-16 computed the exact row drift beyond the model cutoff, moved the source-row boundary to the classical square-root wall, and isolated both a factorable wedge and a distinguished phase coefficient [9]. Its symmetric packet has the form

$$\mathcal{H}_{h,R;U,V}^{\text{HB}}(X) = \sum_{\substack{\ell > U \\ k > V}} \Lambda(\ell) \beta_V(k) r_R(\ell k + h) W(\ell k/X), \quad \beta_V(k) = \sum_{\substack{d|k \\ d \leq V}} \mu(d), \quad (4)$$

with $U = V = \lfloor R^{1/2} \rfloor$. Opening $k = dj$ makes the progression modulus ℓd . TPC-16 controlled one smooth dyadic D -slice of this expansion. A single slice, however, does not yet say that the sum of all smaller divisor labels is controlled.

The first purpose of this paper is to perform that assembly exactly. For $D_0 \leq V$, put

$$\beta_{\leq D_0}(k) = \sum_{\substack{d|k \\ d \leq D_0}} \mu(d), \quad \beta_{(D_0,V]}(k) = \sum_{\substack{d|k \\ D_0 < d \leq V}} \mu(d). \quad (5)$$

Then $\beta_V = \beta_{\leq D_0} + \beta_{(D_0,V]}$ exactly. The new question is whether the complete first term, not merely one of its dyadic components, gives a negligible residual packet.

1.1 Two analytic layers

There are deliberately two distribution layers.

The first layer uses only Maynard's formally published fixed-residue theorem [5, Theorem 1.1]. It yields the three conditions

$$D_0 L^2 \leq X^{1-\eta}, \quad D_0^{12} L^7 \leq X^{4-\eta}, \quad D_0^{20} L^{19} \leq X^{10-\eta}. \quad (6)$$

Under the geometric and finite-row leakage conditions stated later, the whole $d \leq D_0$ packet is $O_A(X(\log X)^{-A})$. In particular, near

$$L = X^{10/21-o(1)}, \quad D_0 = X^{1/21+o(1)}, \quad K = X^{11/21+o(1)}, \quad (7)$$

all divisor slices up to D_0 are assembled. This gives a direct sharp-prefix formulation of the full cumulative region, together with the exact tail normal form proved below. It does not move the published exponent frontier.

The second layer imports a newer signed bilinear theorem of Li [4, Theorem 1.1]. The cited input is explicitly locked to arXiv:2602.20917v6, revised 27 May 2026, and is currently unrefereed. It gives the shorter profile

$$D_0 L^2 \leq X^{1-\eta}, \quad D_0^{12} L^7 \leq X^{4-\eta}. \quad (8)$$

The two bounding lines meet at

$$(\log_X L, \log_X D_0) = \left(\frac{8}{17}, \frac{1}{17} \right), \quad \log_X(LD_0) = \frac{9}{17}. \quad (9)$$

We prove the transfer from Li's signed prime-counting statement to the present smooth von Mangoldt rows. The essential structural identity is that the target weight depends on ℓ but not independently on d . The preprint-dependent theorem is segregated from the published core: if the cited preprint is revised, the Maynard result and all spectral theorems in this paper remain unaffected.

Neither layer estimates the tail $\beta_{(D_0, V]}$. The conclusion is an exact compression of the unresolved coefficient, not a positive lower bound and not a parity-breaking step.

1.2 The phase and dispersion layer

The square-root normal form of TPC-16 reduces the remaining total gate to dyadic blocks

$$\mathcal{H}_{L,K}(\alpha, \beta) = \sum_{\ell, k} c_{\ell, k} e(\alpha \ell + \beta k), \quad LK \asymp_W X, \quad (10)$$

whose arithmetic value is $\mathcal{H}_{L,K}(0, 0)$. We first replace an $X^{1+\varepsilon}$ energy ledger by the explicit estimate

$$E_{L,K} := \sum_{\ell, k} |c_{\ell, k}|^2 \ll_{h, W, \delta} X(\log X)^3. \quad (11)$$

Exact finite Fourier sampling then implies that, at natural height ηX , at most $O_\eta((\log X)^3)$ Nyquist phases can be large. This does not exclude the constant character $(0, 0)$.

A flat-top reproducing kernel makes the obstruction local. If $|\mathcal{H}_{L,K}(0, 0)| \geq \eta X$, then a phase cell of dimensions

$$\frac{(\log X)^B}{L} \times \frac{(\log X)^B}{K} \quad (12)$$

around the origin carries energy $\gg_\eta X$, provided both radii are smaller than the circle scale. If one factor length is at most polylogarithmic, that coordinate is the whole circle and the

statement is one-coordinate localization. Thus a large zero mode is not an isolated measure-zero accident; it is an uncertainty-scale concentration.

Finally, TT^* gives an arithmetic normal form. A large block forces a positive off-diagonal correlation

$$\sum_{\ell_1 \neq \ell_2} \Lambda(\ell_1) \Lambda(\ell_2) \sum_k r_R(\ell_1 k + h) r_R(\ell_2 k + h) \mathcal{W}_{\ell_1, \ell_2, k} \quad (13)$$

of size $\gg XL$ at natural block scale, or $\gg XL/(\log X)^2$ at the dyadic witness scale coming from a failed total estimate. This is weaker than demanding a small operator norm for the whole residual matrix and identifies the next specific arithmetic correlation.

1.3 Claim and dependency ledger

Result	External analytic input	Status if Li v6 changes
Assembled prefix near 11/21	Maynard 2025, Theorem 1.1	unchanged
Assembled prefix near 9/17	Li arXiv v6, Theorem 1.1	must be rechecked
Finite row drift and leakage	TPC-16 and elementary estimates	unchanged
Explicit phase energy	TPC-16 residual energy law	unchanged
Nyquist and uncertainty-cell theorems	finite Fourier analysis	unchanged
TT^* residual-pair reduction	Cauchy–Schwarz and expansion	unchanged

The paper does *not* prove cancellation of $\beta_{(D_0, V]}$, $o(X)$ for every dyadic block, a fixed- h Hardy–Littlewood asymptotic, a new prime-distribution theorem, or a spectral realization of primes or zeta zeros. The translation spectrum used below is the ordinary finite Fourier spectrum of the factor coordinates.

1.4 Organization

[Section 2](#) records the TPC-16 interface. [Section 3](#) proves the one-factor-dependent smooth transfer. [Section 4](#) proves the cumulative prefix theorem. [Sections 5](#) and [6](#) give the published and version-locked exponent profiles. [Section 7](#) derives the explicit phase ledger and finite spectrum. [Section 8](#) proves uncertainty-scale concentration and the TT^* reduction. [Section 9](#) records random-mask cancellation and the sharp abstract no-go model. The finite certificate and next gate appear in [section 10](#).

2 The TPC-16 interface

Throughout, $h \neq 0$ is fixed, $0 < \delta < 1/2$, and

$$R = \lfloor X^{1/2-\delta} \rfloor, \quad U = V = \lfloor R^{1/2} \rfloor. \quad (14)$$

The finite Heath–Brown Ramanujan model is

$$\Lambda_R(n) = \sum_{q \leq R} \frac{\mu(q)}{\varphi(q)} c_q(n), \quad r_R(n) = \Lambda(n) - \Lambda_R(n), \quad (15)$$

where

$$c_q(n) = \sum_{\substack{a \pmod{q} \\ (a, q) = 1}} e(an/q). \quad (16)$$

All constants may depend on h , δ , and fixed smooth weights. An estimate with subscript A is asserted pointwise for each fixed $A > 0$.

2.1 Exact row calibration

For every modulus $m \geq 1$, define the prime progression density and the finite-model complete-period mean by

$$\rho_h(m) = \mathbf{1}_{(m,h)=1} \frac{m}{\varphi(m)}, \quad \rho_{h,R}(m) = \sum_{\substack{q|m \\ q \leq R}} \frac{\mu(q)c_q(h)}{\varphi(q)}. \quad (17)$$

TPC-16 proves the exact drift identity

$$\Delta_{h,R}(m) := \rho_h(m) - \rho_{h,R}(m) = \sum_{\substack{q|m \\ q > R}} \frac{\mu(q)c_q(h)}{\varphi(q)}. \quad (18)$$

For a uniformly bounded compactly supported C^1 family F , its finite-row decomposition is

$$\sum_{j \geq 1} r_R(mj + h)F(mj/X) = \Delta_{h,R}(m) \frac{X}{m} I_0(F) + \mathcal{E}_{m,h}(X; F) + O_F(R(\log(2R))^2), \quad (19)$$

$$\mathcal{E}_{m,h}(X; F) := \sum_{j \geq 1} \Lambda(mj + h)F(mj/X) - \rho_h(m) \frac{X}{m} I_0(F). \quad (20)$$

Thus the residual row has three distinct pieces: prime-progression error, explicit cutoff drift, and finite-model lattice leakage.

The divisor expansion from TPC-16 also gives, on $n \asymp X$,

$$|\Lambda_R(n)| + |r_R(n)| \ll_\varepsilon X^\varepsilon. \quad (21)$$

This deliberately soft pointwise bound is enough for disposing of source and target prime powers.

2.2 Symmetric packet and divisor labels

With β_V as in (4), TPC-16 gives

$$\mathcal{P}_h(X; W) = \mathfrak{S}(h)X I_0(W) - \mathcal{H}_{h,R;U,V}^{\text{HB}}(X) + O_A(X(\log X)^{-A}). \quad (22)$$

Opening $k = dj$ in the packet gives

$$\mathcal{H}_{h,R;U,V}^{\text{HB}}(X) = \sum_{\substack{\ell > U, \\ d \leq V, \\ dj > V}} \Lambda(\ell) \mu(d) r_R(\ell dj + h) W(\ell dj/X). \quad (23)$$

The progression modulus in the inner j -row is $m = \ell d$.

Choose a nonnegative $\psi \in C_c^\infty([1/2, 2])$ such that

$$\sum_{a \in \mathbb{Z}} \psi(x/2^a) = 1 \quad (x > 0). \quad (24)$$

For dyadic L, K , put

$$\mathcal{H}_{L,K}^{\text{HB}} = \sum_{\substack{\ell > U \\ k > V}} \Lambda(\ell) \beta_V(k) r_R(\ell k + h) W(\ell k/X) \psi(\ell/L) \psi(k/K). \quad (25)$$

If $\text{supp } W \subset [a, b] \Subset (0, \infty)$, every nonzero block obeys

$$\frac{aX}{4} \leq LK \leq 4bX. \quad (26)$$

There are $O_W(\log X)$ relevant pairs (L, K) .

For $D_0 \leq V$, define the assembled prefix and tail blocks

$$\mathcal{C}_{L,K}(D_0) := \sum_{\substack{\ell > U \\ k > V}} \Lambda(\ell) \beta_{\leq D_0}(k) r_R(\ell k + h) W(\ell k / X) \psi(\ell / L) \psi(k / K), \quad (27)$$

$$\mathcal{T}_{L,K}(D_0, V) := \sum_{\substack{\ell > U \\ k > V}} \Lambda(\ell) \beta_{(D_0, V]}(k) r_R(\ell k + h) W(\ell k / X) \psi(\ell / L) \psi(k / K). \quad (28)$$

The coefficient identity (5) gives the exact decomposition

$$\mathcal{H}_{L,K}^{\text{HB}} = \mathcal{C}_{L,K}(D_0) + \mathcal{T}_{L,K}(D_0, V). \quad (29)$$

If $L > 2U$ and $K > 2V$, the lower restrictions in (27) are automatic on the support of the dyadic weights. Opening the prefix gives

$$\mathcal{C}_{L,K}(D_0) = \sum_{\ell, d, j \geq 1} \Lambda(\ell) \mu(d) \mathbf{1}_{d \leq D_0} \psi(\ell / L) \psi(dj / K) r_R(\ell dj + h) W(\ell dj / X). \quad (30)$$

The assembled theorem below will assume $L > 2R$. It follows that every row modulus ℓd in (30) exceeds R . Thus the packet is genuinely beyond the finite-model cutoff, and the simple drift calculation of TPC-16 applies without a hidden $\ell d \leq R$ component.

2.3 Square-root and residual-energy interfaces

TPC-16 also proves the asymmetric square-root normal form

$$\mathcal{P}_h(X; W) = \mathfrak{S}(h) X I_0(W) - \mathcal{H}_{h,R,M}^\sharp(X) + O_A(X (\log X)^{-A}), \quad (31)$$

where

$$M = X^{1/2} \exp(-\sqrt{\log X}), \quad \mathcal{H}_{h,R,M}^\sharp(X) = \sum_{\substack{\ell > M \\ k > 1}} \Lambda(\ell) r_R(\ell k + h) W(\ell k / X). \quad (32)$$

The spectral analysis in sections 7 and 8 uses this simpler packet, not the symmetric coefficient β_V .

Finally, for nonnegative $F \in C_c^\infty((0, \infty))$, TPC-16 gives

$$\sum_n r_R(n)^2 F(n/X) = X \log(X/R) I_0(F) + O_{F,\delta}(X). \quad (33)$$

Changing $F(u)$ to $F(u - h/X)$ preserves a uniformly bounded smooth family. Hence for fixed h ,

$$\sum_n r_R(n + h)^2 F(n/X) \ll_{h,F,\delta} X \log X. \quad (34)$$

3 Bilinear distribution inputs and a one-factor smooth transfer

Write

$$E_\pi(t; q, h) = \pi(t; q, h) - \frac{\pi(t)}{\varphi(q)} \quad ((q, h) = 1). \quad (35)$$

The two imported theorems used below have different logical strengths. Maynard controls the absolute double sum in a three-inequality region; Li controls a signed bilinear sum in a larger two-inequality region. We first formulate the common consequence actually needed by the assembled packet.

Definition 3.1 (Signed bilinear distribution profile). Fix $\eta > 0$. A set $\mathfrak{R}_\eta$ of pairs (L, D) has property BD_h if, for every fixed $A, C > 0$, every pair of sequences a_ℓ, b_d satisfying

$$|a_\ell| \leq (\log X)^C, \quad |b_d| \leq (\log X)^C,$$

and every $t \asymp X$, one has

$$\sum_{\substack{\ell \leq 2L, d \leq D \\ (\ell d, h)=1}} a_\ell b_d E_\pi(t; \ell d, h) \ll_{A, C, h, \eta} t (\log t)^{-A} \quad (36)$$

uniformly for $(L, D) \in \mathfrak{R}_\eta$. The profile is understood with a fixed power margin, so replacing X by $t \asymp X$ is harmless.

The log-bounded formulation is deliberate. It is the common consequence of the two imported inputs needed by this paper. Li's signed theorem permits the stronger class of divisor-bounded coefficients, whereas Maynard's absolute unweighted estimate does not by itself permit arbitrary pointwise divisor weights.

3.1 The published and version-locked profiles

Maynard's theorem [5, Theorem 1.1] implies BD_h on

$$\mathfrak{R}_\eta^{\text{M}} = \left\{ (L, D) : DL^2 \leq X^{1-\eta}, \quad D^{12}L^7 \leq X^{4-\eta}, \quad D^{20}L^{19} \leq X^{10-\eta} \right\}. \quad (37)$$

Here the orientation in Maynard's notation is $(Q_1, Q_2) = (D, 2L)$; its three conditions are asymmetric. Indeed, his theorem bounds the sum of the absolute values of $E_\pi(t; d\ell, h)$ over the two factor ranges. Hence log-bounded separated coefficients can be inserted after requesting additional logarithmic saving.

The newer input is the following exact version of the result used here.

Imported input 3.2 (Li v6 bilinear theorem). Theorem 1.1 in version 6 of Li [4] states that, for fixed $h \neq 0$ and arbitrary divisor-bounded complex sequences a_{q_1}, b_{q_2} ,

$$\sum_{\substack{q_1 \leq Q_1, q_2 \leq Q_2 \\ (q_1 q_2, h)=1}} a_{q_1} b_{q_2} E_\pi(t; q_1 q_2, h) \ll_A t (\log t)^{-A} \quad (38)$$

provided, with a fixed positive margin,

$$Q_1^2 Q_2 \leq t^{1-\eta}, \quad Q_1^7 Q_2^{12} \leq t^{4-\eta}. \quad (39)$$

The cited document is an unrefereed preprint revised 27 May 2026; no part of its proof is reproduced or independently certified here.

For Li the orientation is instead $(Q_1, Q_2) = (2L, D)$. Absorbing fixed factors into the power margin gives BD_h on

$$\mathfrak{R}_\eta^{\text{L}} = \left\{ (L, D) : DL^2 \leq X^{1-\eta}, \quad D^{12}L^7 \leq X^{4-\eta} \right\}. \quad (40)$$

The superscript L always denotes the version-locked Li input, not a theorem proved in this paper.

Remark 3.3 (Why other large-modulus theorems are not inserted). Maynard's 3/5 well-factorable theorem [6] applies to collapsed modulus weights possessing triple well-factorability. The specific coefficient $\Lambda(\ell)\mu(d)$ with an independently resolved source factor has no such property in general. Replacing it by a factorable sieve weight would change the exact Vaughan packet. Likewise, splitting a general d into several independent factors fails for a divisor label that is itself a large prime. We therefore do not promote a well-factorable or trilinear theorem to the present packet.

3.2 Smooth von Mangoldt transfer

The signed nature of (38) requires the test weight to preserve separated coefficients. The following lemma records the exact allowable dependence.

Lemma 3.4 (One-factor-dependent smooth transfer). *Let (L, D) belong to a BD_h profile. For every $\ell \leq 2L$, let $F_{\ell, X} \in C_c^1((0, \infty))$ belong to a family with a common compact support and a common C^1 bound. Let $|a_\ell|, |b_d| \leq (\log X)^C$. Then*

$$\sum_{\substack{\ell \leq 2L, d \leq D \\ (\ell d, h)=1}} a_\ell b_d \left[\sum_{j \geq 1} \Lambda(\ell dj + h) F_{\ell, X}(\ell dj/X) - \frac{X}{\varphi(\ell d)} I_0(F_{\ell, X}) \right] \ll_{A, C, h, \eta, \mathcal{F}} X(\log X)^{-A} \quad (41)$$

for every fixed $A > 0$, where $\mathcal{F}$ denotes the common support and C^1 bounds of the weight family.

Proof. We give the transfer in detail because the one-factor restriction is essential. First retain only target primes. Put

$$\omega_{\ell, X}(t) = (\log t) F_{\ell, X}((t - h)/X). \quad (42)$$

The support lies in one fixed interval $t \asymp X$, and

$$|\omega'_{\ell, X}(t)| \ll \frac{\log X}{X} \quad (43)$$

uniformly in ℓ . Stieltjes partial summation makes the difference between the prime progression sum and its global-prime counterpart equal to

$$- \int E_\pi(t; \ell d, h) \omega'_{\ell, X}(t) dt. \quad (44)$$

There are no endpoint terms after extending the compactly supported weight by zero.

After summing (44) over ℓ, d , fix t inside the integral. The coefficient

$$a_{\ell, t} = a_\ell \omega'_{\ell, X}(t) \quad (45)$$

still depends only on ℓ , while b_d depends only on d . Write

$$a_{\ell, t} = \frac{\log X}{X} \tilde{a}_{\ell, t}, \quad |\tilde{a}_{\ell, t}| \leq (\log X)^{C'}$$

using (43). The BD_h estimate applies pointwise to $\tilde{a}_{\ell, t}$ and b_d , uniformly in t . Its $t \asymp X$ scale is multiplied by $(\log X)/X$; integrating over an interval of length $O(X)$ restores the final X scale and loses only one fixed logarithmic power. Increasing the requested saving absorbs that loss.

The global prime sum satisfies, uniformly over the same family,

$$\sum_p (\log p) F_{\ell, X}((p - h)/X) = X I_0(F_{\ell, X}) + O(X \exp(-c\sqrt{\log X})). \quad (46)$$

The accumulated error is negligible because standard divisor estimates give

$$\sum_{\ell \leq 2L, d \leq D} \frac{|a_\ell b_d|}{\varphi(\ell d)} \ll_C (\log X)^{C'}. \quad (47)$$

It remains to restore target prime powers. If $p^\nu \asymp X$, $\nu \geq 2$, and $p^\nu \equiv h \pmod{\ell d}$, then $\ell d \mid p^\nu - h$. Hence the total number of ordered pairs is bounded by a fixed divisor function, and

$$\sum_{\substack{p^\nu \asymp X \\ \nu \geq 2}} (\log p) \tau_3(|p^\nu - h|)^{C'} \ll_{C', \varepsilon} X^{1/2+\varepsilon}. \quad (48)$$

This is absorbed by any prescribed logarithmic saving. Combining the prime, global-main-term, and prime-power estimates proves (41). $\square$

Remark 3.5 (Forbidden pair dependence). The proof does not allow an arbitrary family $F_{\ell,d,X}$. In the signed Li layer, the Stieltjes derivative of such a family would create a genuinely pair-dependent coefficient, outside (38). The packet used below is admissible because its weight has the exact form

$$F_{\ell,X}(u) = W(u)\psi\left(\frac{Xu}{\ell K}\right), \quad F_{\ell,X}(\ell dj/X) = W(\ell dj/X)\psi(dj/K), \quad (49)$$

which is independent of d once ℓ is fixed. The product ledger $LK \asymp_W X$ supplies a common compact support and common C^1 bound.

4 The assembled-prefix theorem

We now pass from one dyadic divisor label to a complete initial segment. We give a direct sharp-prefix proof: the coefficient $\mathbf{1}_{d \leq D_0}$ is inserted into the bilinear distribution theorem itself. This packages the cumulative statement without endpoint loss or dependence on a choice of dyadic partition.

Theorem 4.1 (Cumulative divisor-prefix cancellation). *Fix $h \neq 0$, $0 < \delta < 1/2$, and $\eta > 0$. Let L, K, D_0 satisfy*

$$LK \asymp_W X, \quad L > 2R, \quad 1 \leq D_0 < R, \quad D_0 \leq V, \quad K > 2V, \quad (50)$$

$$(L, D_0) \in \mathfrak{R}_\eta, \quad LD_0 R \leq X^{1-\eta}, \quad (51)$$

where $\mathfrak{R}_\eta$ is any signed bilinear distribution profile with property BD_h . Then, for every $A > 0$,

$$\mathcal{C}_{L,K}(D_0) \ll_{A,h,W,\psi,\delta,\eta} X(\log X)^{-A}. \quad (52)$$

Proof. Start from the opened expression (30). We first remove source prime powers $\ell = p^a$, $a \geq 2$. On the support, $j \asymp_W K/d$, and (21) gives, for every fixed $\varepsilon > 0$,

$$\sum_{\substack{p^a \asymp L \\ a \geq 2}} \Lambda(p^a) \sum_{d \leq D_0} \sum_{j \asymp K/d} |r_R(p^a dj + h)| \\ \ll_{W,\varepsilon} L^{1/2+\varepsilon} K(\log(2D_0))X^\varepsilon \ll_{W,\varepsilon} X^{1+2\varepsilon} L^{-1/2} \log X. \quad (53)$$

Put $\rho_0 = 1/2 - \delta > 0$. Since $L > 2R = X^{\rho_0+o(1)}$, choosing $\varepsilon < \rho_0/8$ makes the last quantity $O(X^{1-\rho_0/4})$. Thus it has a fixed power saving. We may therefore restrict ℓ to primes. In particular, $\ell > R > |h|$ for all sufficiently large X .

For fixed ℓ, d , apply the row decomposition (19) with

$$F_{\ell,X}(u) = W(u)\psi\left(\frac{Xu}{\ell K}\right). \quad (54)$$

The family is uniformly supported and uniformly C^1 by $LK \asymp_W X$, and

$$F_{\ell,X}(\ell dj/X) = W(\ell dj/X)\psi(dj/K).$$

On the reduced rows $(\ell d, h) = 1$, the aggregate prime-progression error is $O_A(X(\log X)^{-A})$ by lemma 3.4, with

$$a_\ell = \Lambda(\ell)\psi(\ell/L)\mathbf{1}_{\ell \text{ prime}}, \quad b_d = \mu(d)\mathbf{1}_{d \leq D_0}. \quad (55)$$

The logarithm in a_ℓ is absorbed by requesting one additional logarithmic power.

It remains to check the nonreduced rows. Since $\ell > |h|$ is prime, $(\ell d, h) > 1$ implies $(d, h) > 1$. If $p \mid (d, h)$, every integer $\ell dj + h$ is divisible by p ; a nonzero von Mangoldt value therefore forces

$$\ell dj + h = p^\nu. \quad (56)$$

For each of the finitely many $p \mid h$, only $O_{h,W}(1)$ exponents occur at scale X , while the number of ordered factorizations $\ell d \mid p^\nu - h$ is $O_{h,\varepsilon}(X^\varepsilon)$. Hence the total prime part of the nonreduced rows is $O_{h,W,\varepsilon}(X^\varepsilon)$. Their continuous prime mean and complete finite-model mean both vanish. Indeed, $\mu(d) \neq 0$ makes d squarefree, every divisor of d lies below R , and the local factor at $p \mid (d, h)$ is $1 - 1 = 0$.

The finite-row lattice leakage in (19) is $O(R(\log(2R))^2)$ per ordered pair (ℓ, d) . Summing the outer weights gives

$$\ll LD_0 R(\log X)^3 \ll X^{1-\eta}(\log X)^3. \quad (57)$$

Finally consider the cutoff drift. For prime $\ell > R$, squarefree $d < R$, and $\ell \nmid d$, all divisors of ℓd below R divide d . Consequently

$$\Delta_{h,R}(\ell d) = \begin{cases} \frac{d}{\varphi(d)(\ell - 1)}, & (d, h) = 1, \\ 0, & (d, h) > 1. \end{cases} \quad (58)$$

After insertion into the continuous term of (19), its total absolute value is

$$\begin{aligned} &\ll_W X \sum_{\ell \lesssim L} \frac{\Lambda(\ell)}{\ell(\ell - 1)} \sum_{d \leq D_0} \frac{\mu(d)^2}{\varphi(d)} \\ &\ll_W \frac{X}{L} (\log(2D_0))^2. \end{aligned} \quad (59)$$

This again has a fixed power saving because $L > 2R$. The source powers, reduced prime errors, nonreduced rows, leakage, and drift are therefore all $O_A(X(\log X)^{-A})$, after shrinking fixed power margins if necessary. $\square$

The conclusion has an exact and useful tail formulation.

Corollary 4.2 (Exact tail normal form). *Under the hypotheses of [theorem 4.1](#),*

$$\mathcal{H}_{L,K}^{\text{HB}} = \mathcal{T}_{L,K}(D_0, V) + O_{A,h,W,\psi,\delta,\eta}(X(\log X)^{-A}) \quad (60)$$

for every $A > 0$.

Proof. Combine the exact coefficient split (29) with (52). $\square$

Remark 4.3 (What the normal form does and does not do). The tail coefficient

$$\beta_{(D_0, V]}(k) = \sum_{\substack{d \mid k \\ D_0 < d \leq V}} \mu(d)$$

is signed. Equation (60) compresses the unknown part to a shorter divisor interval, but supplies neither positivity nor cancellation inside that interval. In particular, it is not a prime-pair lower bound and does not evade the parity obstruction.

5 The published Maynard core

We first specialize the abstract assembly theorem using only the formally published distribution input. Write

$$\lambda = \log_X L, \quad \nu = \log_X D_0. \quad (61)$$

Ignoring fixed power margins, the Maynard region is the polytope

$$2\lambda + \nu \leq 1, \quad 7\lambda + 12\nu \leq 4, \quad 19\lambda + 20\nu \leq 10. \quad (62)$$

The first and third faces meet at

$$(\lambda, \nu) = \left(\frac{10}{21}, \frac{1}{21} \right), \quad \lambda + \nu = \frac{11}{21}, \quad (63)$$

and the second inequality is then strict.

Theorem 5.1 (Published assembled wedge). *Fix $h \neq 0$, $0 < \delta < 1/2$, and $\eta > 0$. Suppose (50) and the finite-row leakage condition in (51) hold, together with*

$$D_0 L^2 \leq X^{1-\eta}, \quad D_0^{12} L^7 \leq X^{4-\eta}, \quad D_0^{20} L^{19} \leq X^{10-\eta}. \quad (64)$$

Then, for every $A > 0$,

$$\mathcal{C}_{L,K}(D_0) \ll_A X(\log X)^{-A}, \quad \mathcal{H}_{L,K}^{\text{HB}} = \mathcal{T}_{L,K}(D_0, V) + O_A(X(\log X)^{-A}). \quad (65)$$

All implied constants may depend on the fixed data listed in [theorem 4.1](#).

Proof. The three inequalities place (L, D_0) in $\mathfrak{R}_\eta^{\text{M}}$. Apply [theorem 4.1](#) and [corollary 4.2](#). $\square$

The following explicit family reaches the vertex from the interior.

Corollary 5.2 (A complete prefix near the 11/21 boundary). *Fix $0 < \sigma < 1/100$ and*

$$\frac{1}{42} + 3\sigma < \delta < \frac{1}{4}. \quad (66)$$

Set, up to harmless integer parts,

$$L = X^{10/21-\sigma}, \quad D_0 = X^{1/21-\sigma}, \quad K = X/L = X^{11/21+\sigma}. \quad (67)$$

Then (65) holds for every $A > 0$. Thus all divisor labels

$$d \leq X^{1/21-\sigma} \quad (68)$$

are removed in one cumulative packet.

Proof. The three monomials in (64) have exponents

$$1 - 3\sigma, \quad \frac{82}{21} - 19\sigma, \quad 10 - 39\sigma. \quad (69)$$

They therefore have fixed power margins. Moreover,

$$LD_0 R = X^{1+1/42-2\sigma-\delta+o(1)} \leq X^{1-5\sigma+o(1)} \quad (70)$$

by (66). The same lower bound on δ gives $L > 2R$ for large X . The upper bound $\delta < 1/4$ gives $D_0 < V$, while $K > 2V$ is immediate. Hence all hypotheses of [theorem 5.1](#) hold after choosing a fixed $\eta > 0$ smaller than the displayed margins. $\square$

Remark 5.3 (The cumulative packaging of TPC-16 slices). TPC-16 proved the corresponding estimate for one smooth divisor slice $d \asymp D$. Under uniform fixed margins, one could also sum $O(\log X)$ such estimates after requesting additional logarithmic saving. The statement here instead inserts $\mu(d)\mathbf{1}_{d \leq D_0}$ itself in one application and records the exact tail normal form. It does not increase the published exponent vertex.

6 A version-locked enlargement from Li v6

This section is logically separate from the published core. It applies the imported input stated in [Imported input 3.2](#); every assertion here must be rechecked if that exact arXiv version changes.

In the exponent coordinates (61), the Li profile is

$$2\lambda + \nu \leq 1, \quad 7\lambda + 12\nu \leq 4. \quad (71)$$

The two faces meet at

$$(\lambda, \nu) = \left(\frac{8}{17}, \frac{1}{17} \right), \quad \lambda + \nu = \frac{9}{17}. \quad (72)$$

At this point the deleted Maynard monomial has exponent

$$19\lambda + 20\nu = \frac{172}{17} > 10. \quad (73)$$

Thus the enlargement is genuine and is not a relabelling of the published region.

Theorem 6.1 (Li-v6-dependent assembled wedge). *Assume [Imported input 3.2](#). Fix $h \neq 0$, $0 < \delta < 1/2$, and $\eta > 0$. Suppose (50), the leakage condition in (51), and*

$$D_0 L^2 \leq X^{1-\eta}, \quad D_0^{12} L^7 \leq X^{4-\eta} \quad (74)$$

hold. Then, for every $A > 0$,

$$\mathcal{C}_{L,K}(D_0) \ll_A X(\log X)^{-A}, \quad \mathcal{H}_{L,K}^{\text{HB}} = \mathcal{T}_{L,K}(D_0, V) + O_A(X(\log X)^{-A}). \quad (75)$$

Proof. The two monomial conditions place (L, D_0) in $\mathfrak{R}_\eta^L$. The assertion follows from [theorem 4.1](#) and [corollary 4.2](#). The smooth-transfer step is valid here specifically because of (49); it does not invoke a pair-dependent coefficient. $\square$

There is a small family of interior approaches to the vertex. The following one keeps the arithmetic transparent.

Corollary 6.2 (An explicit interior family). *Assume [Imported input 3.2](#). Fix*

$$0 < \sigma < \frac{1}{1000}, \quad \frac{1}{34} + 2\sigma < \delta < \frac{1}{4}, \quad (76)$$

and set, up to integer parts,

$$L = X^{8/17-\sigma}, \quad D_0 = X^{1/17+\sigma/2}, \quad K = X/L = X^{9/17+\sigma}. \quad (77)$$

Then (75) holds for every $A > 0$. The assembled modulus product has size

$$LD_0 = X^{9/17-\sigma/2}, \quad (78)$$

which is strictly larger than $X^{11/21}$ throughout the stated σ range.

Proof. The exponents of the two monomials in (74) are exactly

$$1 - \frac{3}{2}\sigma, \quad 4 - \sigma. \quad (79)$$

The finite-row leakage has exponent

$$\log_X(LD_0R) = 1 + \frac{1}{34} - \frac{\sigma}{2} - \delta + o(1) < 1 - \frac{5}{2}\sigma + o(1). \quad (80)$$

Also, $\delta > 1/34 + \sigma$ implies $L > 2R$ for large X . The bounds $D_0 < R$, $D_0 < V$, and $K > 2V$ follow from $\delta < 1/4$. Finally,

$$\frac{9}{17} - \frac{\sigma}{2} - \frac{11}{21} = \frac{2}{357} - \frac{\sigma}{2} > 0 \quad (81)$$

for $\sigma < 1/1000$. $\square$

Remark 6.3 (Status of the extension). The implication from [Imported input 3.2](#) to [theorem 6.1](#) is a proved transfer, not a numerical heuristic. Its external input is nevertheless a currently unrefereed, version-locked preprint. Accordingly, the 9/17 vertex is advertised only inside this segregated layer. Theorems [4.1](#) and [5.1](#), and all Fourier and TT^* statements below, do not depend on Li's result.

7 Explicit phase energy and finite translation spectrum

We now return to the square-root packet [\(32\)](#). This part is independent of both large-modulus inputs. For a relevant dyadic pair (L, K) define

$$c_{\ell k} := \Lambda(\ell) r_R(\ell k + h) W(\ell k/X) \psi(\ell/L) \psi(k/K) \mathbf{1}_{\ell > M} \mathbf{1}_{k > 1}, \quad (82)$$

$$\mathcal{H}_{L,K}(\alpha, \beta) := \sum_{\ell, k} c_{\ell k} e(\alpha \ell + \beta k), \quad (\alpha, \beta) \in \mathbb{T}^2, \quad (83)$$

$$E_{L,K} := \sum_{\ell, k} |c_{\ell k}|^2. \quad (84)$$

The zero phase is precisely the arithmetic dyadic block. Every nonzero block has $LK \asymp_W X$ by [\(26\)](#).

Theorem 7.1 (Explicit block-energy ledger). *For every relevant pair (L, K) ,*

$$E_{L,K} \ll_{h,W,\psi,\delta} X(\log X)^3. \quad (85)$$

Moreover,

$$\int_{\mathbb{T}} |\mathcal{H}_{L,K}(\alpha, 0)|^2 d\alpha \ll K E_{L,K} \ll \frac{X^2(\log X)^3}{L}, \quad (86)$$

$$\int_{\mathbb{T}} |\mathcal{H}_{L,K}(0, \beta)|^2 d\beta \ll L E_{L,K} \ll \frac{X^2(\log X)^3}{K}, \quad (87)$$

$$\int_{\mathbb{T}^2} |\mathcal{H}_{L,K}(\alpha, \beta)|^2 d\alpha d\beta = E_{L,K}. \quad (88)$$

Proof. Group [\(84\)](#) by $n = \ell k$. Since a divisor d of n has $\Lambda(d) \neq 0$ only when $d = p^a$, one has

$$\sum_{d|n} \Lambda(d)^2 = \sum_{p^a || n} a(\log p)^2 \leq (\log n)^2. \quad (89)$$

It follows that

$$\begin{aligned} E_{L,K} &\ll_{\psi} \sum_n r_R(n+h)^2 |W(n/X)|^2 \sum_{\ell|n} \Lambda(\ell)^2 \\ &\ll X(\log X)^3 \end{aligned} \quad (90)$$

by [\(34\)](#).

For the first one-phase identity, Parseval in ℓ and Cauchy–Schwarz in the $O(K)$ possible values of k give

$$\int_{\mathbb{T}} |\mathcal{H}(\alpha, 0)|^2 d\alpha = \sum_{\ell} \left| \sum_k c_{\ell k} \right|^2 \ll K \sum_{\ell, k} |c_{\ell k}|^2.$$

Interchanging the roles of ℓ, k proves [\(87\)](#); two-dimensional Parseval proves [\(88\)](#). Finally use $LK \asymp_W X$. $\square$

The continuum identity has an exact finite counterpart. Choose integers

$$N_L > 4L, \quad N_K > 4K, \quad N_L \asymp L, \quad N_K \asymp K. \quad (91)$$

Theorem 7.2 (Exact Nyquist Parseval and exceptional-phase count). *With (91),*

$$\frac{1}{N_L N_K} \sum_{a=0}^{N_L-1} \sum_{b=0}^{N_K-1} \left| \mathcal{H}_{L,K} \left(\frac{a}{N_L}, \frac{b}{N_K} \right) \right|^2 = E_{L,K}. \quad (92)$$

Consequently, for every $A_0 > 0$,

$$\# \left\{ (a, b) : \left| \mathcal{H}_{L,K} \left(\frac{a}{N_L}, \frac{b}{N_K} \right) \right| \geq A_0 \right\} \leq \frac{N_L N_K E_{L,K}}{A_0^2}. \quad (93)$$

In particular, there are at most

$$O_\eta((\log X)^3) \text{ phases of height at least } \eta X, \quad (94)$$

and at most $O_\eta((\log X)^5)$ phases of height at least $\eta X / \log X$.

Proof. Discrete character orthogonality gives

$$\frac{1}{N_L} \sum_{a=0}^{N_L-1} e(a(\ell_1 - \ell_2)/N_L) = \mathbf{1}_{\ell_1 = \ell_2}$$

on the frequency support, because $|\ell_1 - \ell_2| < N_L$; the analogous identity holds in k . This proves (92). Markov's inequality gives (93). Now use $N_L N_K \asymp LK \asymp_W X$ and (85). $\square$

This sampling identity has a literal finite spectral interpretation. Embed $(c_{\ell k})$ into $\ell^2(\mathbb{Z}/N_L \mathbb{Z} \times \mathbb{Z}/N_K \mathbb{Z})$ and set

$$\Phi_{a,b}(\ell, k) = \frac{1}{\sqrt{N_L N_K}} e \left(\frac{a\ell}{N_L} + \frac{bk}{N_K} \right). \quad (95)$$

These are the joint eigencharacters of the two coordinate translations. With the convention $\langle f, g \rangle = \sum f \bar{g}$, one has

$$|\langle c, \Phi_{a,b} \rangle|^2 = \frac{1}{N_L N_K} \left| \mathcal{H}_{L,K} \left(-\frac{a}{N_L}, -\frac{b}{N_K} \right) \right|^2. \quad (96)$$

Negating (a, b) merely permutes the Nyquist grid. Removing a spectral projection of height at least A_0 therefore decreases Frobenius energy by at least $A_0^2/(N_L N_K)$. At natural height only $O((\log X)^3)$ such increments are possible.

Remark 7.3 (No distinguished-phase conclusion from Parseval). The arithmetic point $(\alpha, \beta) = (0, 0)$ is the constant translation character. Equations (92) and (93) prove that large characters are sparse, but do not say that the constant one is absent. No zeta-zero or prime “eigenvalue” is involved in this finite spectrum.

8 Uncertainty-cell concentration and a TT^* normal form

The preceding energy theorem makes large phases sparse. We next show that a large zero phase must occupy energy at the reciprocal resolution of the two factor coordinates.

Choose $\vartheta \in C_c^\infty((0, \infty))$ equal to 1 on $\text{supp } \psi$, and define the periodic flat-top kernels

$$V_L(\alpha) = \sum_{n \in \mathbb{Z}} \vartheta(n/L) e(n\alpha), \quad V_K(\beta) = \sum_{n \in \mathbb{Z}} \vartheta(n/K) e(n\beta). \quad (97)$$

Poisson summation gives, for every $B_0 > 0$,

$$|V_L(\alpha)| \ll_{B_0, \vartheta} L(1 + L\|\alpha\|)^{-B_0}, \quad (98)$$

and similarly for V_K . For $T \geq 2$, put

$$\Omega_T = \left\{ (\alpha, \beta) \in \mathbb{T}^2 : \|\alpha\| \leq T/L, \quad \|\beta\| \leq T/K \right\}. \quad (99)$$

If one of the displayed radii exceeds $1/2$, the corresponding condition is interpreted as the whole circle.

Theorem 8.1 (Flat-top localization). *For every $B_0 > 0$,*

$$|\mathcal{H}_{L,K}(0,0)| \leq C_{B_0} \sqrt{LK} \left[\left(\int_{\Omega_T} |\mathcal{H}_{L,K}(\alpha, \beta)|^2 d\alpha d\beta \right)^{1/2} + T^{-B_0} E_{L,K}^{1/2} \right]. \quad (100)$$

Consequently, for each fixed $\eta > 0$, one may choose a fixed $B > 0$ such that, with $T = (\log X)^B$,

$$|\mathcal{H}_{L,K}(0,0)| \geq \eta X \implies \int_{\Omega_T} |\mathcal{H}_{L,K}|^2 \gg_{\eta} X, \quad (101)$$

$$|\mathcal{H}_{L,K}(0,0)| \geq \frac{\eta X}{\log X} \implies \int_{\Omega_T} |\mathcal{H}_{L,K}|^2 \gg_{\eta} \frac{X}{(\log X)^2}. \quad (102)$$

Proof. Fourier reproduction gives the exact identity

$$\mathcal{H}_{L,K}(0,0) = \int_{\mathbb{T}^2} \mathcal{H}_{L,K}(\alpha, \beta) V_L(-\alpha) V_K(-\beta) d\alpha d\beta. \quad (103)$$

Indeed, integration selects the frequencies $n = \ell$ and $m = k$, on which both flat-top multipliers equal 1.

On Ω_T , Cauchy–Schwarz and Parseval for the kernels give

$$\left| \int_{\Omega_T} \mathcal{H} V_L V_K \right| \ll \sqrt{LK} \left(\int_{\Omega_T} |\mathcal{H}|^2 \right)^{1/2}.$$

On the complement, apply Cauchy–Schwarz again, use (88), and integrate the rapid decay (98). Taking the decay exponent larger than the requested B_0 yields

$$\left| \int_{\mathbb{T}^2 \setminus \Omega_T} \mathcal{H} V_L V_K \right| \ll_{B_0} \sqrt{LK} T^{-B_0} E_{L,K}^{1/2}.$$

This proves (100).

Now $LK \asymp_W X$ and $E_{L,K}^{1/2} \ll \sqrt{X}(\log X)^{3/2}$. Choose B and then B_0 so that the tail in (100) is smaller than one half of the relevant threshold. Rearranging proves (101) and (102). $\square$

The same obstruction has a more arithmetic formulation. Put

$$S_k = \sum_{\ell} c_{\ell k} \quad (104)$$

and define

$$\begin{aligned} \mathcal{C}_{L,K}^{\text{off}} &:= \sum_{\ell_1 \neq \ell_2} \Lambda(\ell_1) \Lambda(\ell_2) \psi(\ell_1/L) \psi(\ell_2/L) \\ &\quad \times \sum_k r_R(\ell_1 k + h) r_R(\ell_2 k + h) W(\ell_1 k/X) W(\ell_2 k/X) \\ &\quad \times \psi(k/K)^2 \mathbf{1}_{\ell_1, \ell_2 > M} \mathbf{1}_{k > 1}. \end{aligned} \quad (105)$$

All quantities are real because the weights are real.

Theorem 8.2 (TT^* residual-pair reduction). *One has the exact identity*

$$\sum_k |S_k|^2 = E_{L,K} + \mathcal{C}_{L,K}^{\text{off}}. \quad (106)$$

In particular,

$$\mathcal{C}_{L,K}^{\text{off}} = o(XL) \implies \mathcal{H}_{L,K}(0,0) = o(X). \quad (107)$$

Conversely, for every fixed $\eta > 0$,

$$|\mathcal{H}_{L,K}(0,0)| \geq \eta X \implies \mathcal{C}_{L,K}^{\text{off}} \geq (c_W \eta^2 + o(1))XL, \quad (108)$$

$$|\mathcal{H}_{L,K}(0,0)| \geq \frac{\eta X}{\log X} \implies \mathcal{C}_{L,K}^{\text{off}} \gg_{W,\eta} \frac{XL}{(\log X)^2}. \quad (109)$$

Proof. Expanding $\sum_k |S_k|^2$ separates the diagonal $\ell_1 = \ell_2$, which is exactly $E_{L,K}$, from (105). This proves (106). Since only $O(K)$ values of k occur,

$$|\mathcal{H}_{L,K}(0,0)|^2 = \left| \sum_k S_k \right|^2 \ll K(E_{L,K} + \mathcal{C}_{L,K}^{\text{off}}). \quad (110)$$

Now $K \asymp_W X/L$ and

$$E_{L,K} \ll X(\log X)^3 = o(XL), \quad (111)$$

because every nonzero square-root block has $L \gg M = X^{1/2-o(1)}$. The sufficient condition and both converse estimates follow from (110). $\square$

Corollary 8.3 (A failed total gate produces a two-form witness). *Suppose that along a sequence of X ,*

$$|\mathcal{H}_{h,R;M}^\sharp(X)| \geq \eta X \quad (112)$$

for some fixed $\eta > 0$. Then for one of the $O_W(\log X)$ relevant dyadic pairs (L, K) ,

$$|\mathcal{H}_{L,K}(0,0)| \gg_{W,\eta} \frac{X}{\log X} \quad (113)$$

and

$$\mathcal{C}_{L,K}^{\text{off}} \gg_{W,\eta} \frac{XL}{(\log X)^2}. \quad (114)$$

After a further dyadic decomposition of $|\ell_1 - \ell_2|$, one separation scale Δ carries a positive contribution

$$\gg_{W,\eta} \frac{XL}{(\log X)^3}. \quad (115)$$

The witness may be restricted to distinct prime $\ell_1, \ell_2 \asymp L$: if $\mathcal{C}_{L,K}^{\text{off},\mathbb{P}}$ denotes (105) with both source variables restricted to primes, then

$$\mathcal{C}_{L,K}^{\text{off},\mathbb{P}} \gg_{W,\eta} \frac{XL}{(\log X)^2}, \quad (116)$$

and one prime separation scale contributes $\gg_{W,\eta} XL/(\log X)^3$. The determinant of its two affine forms is

$$\det \begin{pmatrix} \ell_1 & h \\ \ell_2 & h \end{pmatrix} = h(\ell_1 - \ell_2). \quad (117)$$

Proof. Insert the dyadic partition in the two factor coordinates. The product constraint leaves $O_W(\log X)$ nonzero pairs, so the triangle inequality gives (113); then apply (109). There are $O(\log L)$ dyadic ranges for the nonzero integer $|\ell_1 - \ell_2|$, so at least one has a positive contribution of the size in (115).

Finally, define the prime-restricted coefficients $c_{\ell k}^{\mathbb{P}} = c_{\ell k} \mathbf{1}_{\ell \text{ prime}}$ and the associated polynomial $\mathcal{H}_{L,K}^{\mathbb{P}}$. The total zero-phase contribution of source prime powers $\ell = p^a$, $a \geq 2$, is

$$\ll_{W,\varepsilon} XL^{-1/2+\varepsilon} = o(X/\log X), \quad (118)$$

by (21) and $L \gg M$. Removing it before the TT^* expansion gives

$$|\mathcal{H}_{L,K}^{\mathbb{P}}(0,0)| \gg_{W,\eta} X/\log X.$$

Now repeat the proof of theorem 8.2 for $c^{\mathbb{P}}$; its diagonal energy is at most $E_{L,K}$. This proves (116). Decomposing this prime-restricted correlation by $|\ell_1 - \ell_2|$ proves the stated prime separation-scale bound. Formula (117) is immediate. $\square$

Remark 8.4 (The next arithmetic gate). The target in (105) is weaker than a small operator norm for the whole residual matrix: only the source-prime direction and its TT^* self-correlation matter. A future theorem showing $o(XL)$ for this averaged two-form correlation would close the corresponding zero-phase block. No such estimate is proved here.

9 Random masks and a sharp information boundary

The zero-phase obstruction is deterministic. Independent signs erase it uniformly over all phases at a much smaller scale, which separates a generic entropy statement from the arithmetic alignment that remains.

Theorem 9.1 (Uniform random-entry cancellation). *Let $(\varepsilon_{\ell k})$ be independent Rademacher variables and put*

$$\mathcal{H}_{L,K}^{\varepsilon}(\alpha, \beta) = \sum_{\ell, k} \varepsilon_{\ell k} c_{\ell k} e(\alpha \ell + \beta k). \quad (119)$$

There is an absolute $C > 0$ such that, for $u \geq 0$,

$$\mathbb{P} \left(\sup_{\alpha, \beta \in \mathbb{T}} |\mathcal{H}_{L,K}^{\varepsilon}(\alpha, \beta)| > C \sqrt{E_{L,K}(\log(2LK) + u)} \right) \leq 2e^{-u}. \quad (120)$$

In particular, with probability $1 - O(X^{-10})$,

$$\sup_{\alpha, \beta} |\mathcal{H}_{L,K}^{\varepsilon}(\alpha, \beta)| \ll \sqrt{X}(\log X)^2 = o(X). \quad (121)$$

Proof. For each fixed phase, the Rademacher exponential-moment inequality gives

$$\mathbb{P}(|\mathcal{H}^{\varepsilon}(\alpha, \beta)| > t) \leq 2 \exp(-ct^2/E_{L,K}). \quad (122)$$

A rectangular grid with $O(LK)$ points and fixed oversampling controls the supremum of a trigonometric polynomial with degrees $O(L)$ and $O(K)$ up to an absolute constant; this follows, for example, from the Bernstein derivative inequality in each coordinate. Apply (122) on that grid and take a union bound. This proves (120). Choosing $u = 12 \log X$ and using (85) proves (121). This is the standard Salem–Zygmund net argument; see Kahane [3], Zygmund [10]. $\square$

Even one sign per source row is enough on the zero- β slice, and hence at the distinguished zero phase. Let

$$A_{\ell} = \sum_k c_{\ell k}. \quad (123)$$

Corollary 9.2 (Uniform random-row cancellation). *For independent Rademacher signs (ε_ℓ) , with probability $1 - O(X^{-10})$,*

$$\sup_{\alpha \in \mathbb{T}} \left| \sum_{\ell} \varepsilon_\ell A_\ell e(\alpha \ell) \right| \ll \frac{X(\log X)^2}{\sqrt{L}} \leq X^{3/4+o(1)} = o(X). \quad (124)$$

Proof. By (86),

$$\sum_{\ell} |A_\ell|^2 = \int_{\mathbb{T}} |\mathcal{H}_{L,K}(\alpha, 0)|^2 d\alpha \ll \frac{X^2(\log X)^3}{L}. \quad (125)$$

The one-dimensional version of the preceding net argument contributes one further factor $\sqrt{\log X}$. Finally, every nonzero square-root block has $L \gg M = X^{1/2-o(1)}$. $\square$

Randomization is not available in the arithmetic packet. The next proposition shows that this distinction is logically necessary: the phase-energy ledgers proved so far are compatible with a macroscopic deterministic zero mode.

Proposition 9.3 (Sharp abstract countermodel). *Fix a sequence of product blocks with $LK \asymp X$ and $\min(L, K) \rightarrow \infty$. Suppose there are intervals of lengths $\asymp L$ and $\asymp K$ whose product rectangle lies in a region on which $|W|$ and both dyadic weights are uniformly bounded below. There exists a real sequence $r_X^*(m)$ such that*

- (i) $|r_X^*(m)| \ll \sqrt{\log X} \ll_\varepsilon X^\varepsilon$;
- (ii) *its ambient energy on a fixed interval of length $\asymp X$ is $\asymp X \log X$;*
- (iii) *the corresponding block satisfies all upper bounds in [theorems 7.1](#) and [7.2](#);*
- (iv) *nevertheless,*

$$|\mathcal{H}_{L,K}^*(0, 0)| \asymp X. \quad (126)$$

This is an information-theoretic countermodel, not a statement about the arithmetic residual r_R .

Proof. Let $\mathcal{A}$ be the set of target coordinates $m = \ell k + h$ occurring in the block with $\Lambda(\ell) \neq 0$. On these coordinates define

$$r_X^*(m) = \operatorname{sgn} W((m - h)/X), \quad (127)$$

with an arbitrary choice at zeros of W . Then

$$\mathcal{H}_{L,K}^*(0, 0) = \sum_{\ell, k} \Lambda(\ell) |W(\ell k/X)| \psi(\ell/L) \psi(k/K). \quad (128)$$

After restricting to fixed subintervals on which the weights do not vanish, the prime number theorem in the ℓ coordinate gives (126).

On the block, $|r_X^*| = 1$, so

$$E_{L,K}^* \ll K \sum_{\ell \asymp L} \Lambda(\ell)^2 \ll X \log X. \quad (129)$$

All one- and two-phase upper bounds then follow from the same Parseval and Cauchy–Schwarz arguments as before; exact Nyquist Parseval is purely algebraic.

The number of distinct block coordinates is at most the number of ordered pairs, hence

$$|\mathcal{A}| \ll K \#\{\ell \asymp L : \Lambda(\ell) \neq 0\} \leq K(\pi(2L) + O(\sqrt{L} \log L)) \ll \frac{X}{\log L} = o(X). \quad (130)$$

There are therefore $\asymp X$ unused target coordinates in a slightly larger fixed ambient interval. Set $r_X^*(m) = B_X$ on those coordinates, where $B_X \asymp \sqrt{\log X}$ is chosen to give any prescribed positive constant multiple of $X \log X$ ambient energy. This does not change the block and proves all four assertions. $\square$

Remark 9.4 (Exact scope of the no-go statement). The constructed r_X^* is neither $\Lambda - \Lambda_R$ nor claimed to satisfy its divisor identities, progression laws, or local correlations. The proposition proves only that

$$\text{band limitation} + \text{pointwise size} + \text{total energy} + \text{phase mean squares} \quad (131)$$

cannot, by themselves, force cancellation at the distinguished zero phase. New progress must use additional arithmetic structure. The TT^* reduction identifies one precise place for it: the averaged two-form correlation (105).

10 Finite certificate, maze exits, and conclusion

The companion script

`experiments/tpc17_certificate.py`

performs four deterministic checks:

1. it solves the Maynard and Li boundary intersections over exact rational arithmetic and verifies the gap $9/17 - 11/21 = 2/357$;
2. it verifies the explicit monomial and δ ledgers at a sample interior point;
3. it checks $\beta_V = \beta_{\leq D_0} + \beta_{(D_0, V]}$ on a finite range; and
4. it tests exact Nyquist Parseval and the TT^* diagonal/off-diagonal identity on a deterministic finite matrix.

The generated JSON file records all inputs and residual errors. These checks certify algebra and implementation only. They are not evidence for an unproved prime-pair asymptotic and are not used in any proof.

The resulting maze has three mathematically distinct exits.

Route	Target	Honest stopping result
Tail route	Estimate a nontrivial subrange of $\beta_{(D_0, V]}$ by additional factor shape or cancellation	A larger exact prefix, or a theorem identifying which tail shapes are inaccessible to current distribution inputs
Dispersion route	Prove $o(XL)$ for the averaged two-form correlation (105)	A zero-phase cancellation theorem on a restricted corridor, or a sharp lower-bound obstruction for existing TT^* information
Hybrid route	Use divisor-tail structure before the TT^* expansion	A determinant-resolved normal form combining $d > D_0$ with $h(\ell_1 - \ell_2)$, even if the final cancellation remains open

The paper therefore advances on two fronts. Analytically, it packages the factorable divisor slices as a complete initial segment and compresses the unresolved Vaughan coefficient to an exact tail. Spectrally, it turns a possible distinguished zero-mode failure into a sparse uncertainty-cell concentration and then into one explicit off-diagonal two-form correlation. The countermodel proves that the phase mean-square ledgers available here, even together with pointwise and total-energy bounds, do not close that gate.

The next useful theorem must consequently exploit arithmetic structure in either the Möbius tail or the determinant-resolved residual pair. This is a narrower and more falsifiable target than the original fixed-shift correlation, but it remains well short of a twin-prime or general Hardy–Littlewood theorem.

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